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Postoperative pain after emergency pulpectomy with or without photobiomodulation in symptomatic irreversible pulpitis A randomized clinical trial

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To evaluate postoperative pain following emergency pulpectomy in permanent teeth diagnosed with symptomatic irreversible pulpitis and to compare outcomes with or without adjunctive low-level laser therapy (LLLT). This parallel, randomized, single-blind clinical trial included 70 patients with permanent teeth diagnosed with symptomatic irreversible pulpitis. Patients were randomly assigned to two groups ($n = 35$ each): the experimental group received gallium–aluminum–arsenide laser therapy (808 nm, 100 mW), while the control group received no laser intervention. Postoperative pain was assessed using the Numerical Rating Scale (NRS-10) and the Visual Analogue Scale (VAS). Bite sensitivity was evaluated before and after emergency treatment. Demographic characteristics and analgesic intake were recorded. The primary outcome was postoperative spontaneous pain assessed at multiple time points. Secondary outcomes included bite sensitivity and analgesic intake. Time-dependent reductions in postoperative pain were observed in both groups. Inter-group comparisons showed statistically significant differences at isolated time points (12 and 24 h), with higher pain scores in the experimental group; however, these differences did not translate into a significant overall treatment effect. Longitudinal analysis accounting for repeated measures demonstrated a significant effect of time ($p = 0.001$), while treatment group was not a significant determinant of postoperative pain. Effect estimates are presented with 95% confidence intervals. Within the limitations of this pragmatic emergency-care study, adjunctive low-level laser therapy was not associated with a reduction in postoperative pain in permanent teeth with symptomatic irreversible pulpitis.

Keywords Endodontics, Photobiomodulation, Low-level laser therapy, Postoperative pain, Irreversible pulpitis

Postoperative pain is a common complication, reported in approximately 25–69% of patients undergoing endodontic treatment[1,2]. Several factors may contribute to this condition, including trauma caused by inadequate instrumentation, chemical injury to periapical tissues due to high concentrations of irrigating solutions, extrusion of obturating materials, microbial factors, and the persistence of contaminated debris in the periapical region[3–8]. Studies have shown that the incidence of postoperative pain is higher in teeth treated under emergency conditions, which may be attributed to temporary restorations that permit microleakage and bacterial contamination[9,10]. Additionally, in patients undergoing emergency endodontic procedures, remnants

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of vital pulp tissue may remain after pulpectomy, favoring the development of signs and symptoms that negatively affect their quality of life[11].

Chemical mediators of inflammation, such as prostaglandins, increase vascular permeability, enhance the chemotactic activity of immune cells, and sensitize nociceptors, thereby intensifying pain perception[12,13]. Various strategies have been proposed to reduce postoperative pain, including pharmacological interventions, occlusal adjustment, anesthetic techniques, and photobiomodulation therapy[2,3,6,14–17]. Among pharmacological agents, nonsteroidal anti-inflammatory drugs (NSAIDs) are the most commonly prescribed in dentistry, with ibuprofen considered the standard for managing postoperative pain associated with symptomatic irreversible pulpitis[2,15,18–21]. Ibuprofen provides short-term analgesia with minimal adverse effects[4,16]; however, its use may need to be combined with other medications to achieve prolonged and satisfactory pain relief[21].

Photobiomodulation therapy has been proposed as an effective method for reducing postoperative pain, with reported benefits including analgesia, tissue regeneration, and cellular biostimulation. Its mechanisms are thought to involve several biological actions, such as localized surface heating without systemic temperature increase, enhanced circulation and removal of inflammatory mediators, inhibition of pro-inflammatory factors, stimulation of mitochondrial respiration, and the release of neurotransmitters within inflamed tissues[22]. Evidence suggests that photobiomodulation can effectively reduce postoperative pain, serving as a potential alternative to analgesics and anti-inflammatory drugs[23]. The therapy increases local blood flow, decreases prostaglandin E_2 levels, and inhibits cyclooxygenase-2 activity[24]. Importantly, low-level laser therapy does not cause structural changes to cellular tissues; instead, the absorbed light stimulates mitochondrial photoreceptors, triggering beneficial effects that reduce inflammation and pain[25]. Red laser light has been identified as the most effective wavelength for pain control, whereas infrared light induces vasomotor relaxation—both without systemic adverse effects, unlike pharmacologic therapy[26].

Despite the growing interest in photobiomodulation therapy (PBM) for pain control in endodontics, important methodological and biological factors must be considered when interpreting its clinical effectiveness. The existing literature is characterized by substantial heterogeneity in PBM protocols, with wide variations in wavelength, power output, energy density, irradiation time, number of application points, and timing of laser application. Such variability may significantly influence treatment outcomes and partially explain the inconsistent results reported across clinical studies, thereby limiting direct comparison and the establishment of standardized clinical recommendations[27]. In addition to protocol-related factors, symptomatic irreversible pulpitis (SIP) presents a disease-specific inflammatory burden characterized by intense vascular changes, increased intrapulpal pressure, and elevated levels of pro-inflammatory mediators, including prostaglandins and cytokines. This pronounced inflammatory environment leads to enhanced peripheral and central sensitization, which may reduce the effectiveness of adjunctive analgesic strategies such as PBM, particularly when applied during emergency endodontic interventions primarily aimed at pain relief[28].

The objective of this randomized clinical trial was to evaluate postoperative pain outcomes following pulpectomy in permanent teeth with symptomatic irreversible pulpitis and to compare the use of adjunctive low-level laser therapy with conventional emergency treatment, as well as to assess analgesic intake after treatment. The null hypothesis tested was that photobiomodulation therapy is not associated with a reduction in postoperative pain in cases of symptomatic irreversible pulpitis, whereas the alternative hypothesis was that it is associated with lower postoperative pain following pulpectomy.

Methodology

Study design and ethics

This parallel, randomized, double-blind clinical trial was conducted in accordance with the Declaration of Helsinki and all relevant guidelines and regulations, including the Consolidated Standards of Reporting Trials (CONSORT) 2025 guidelines[29]. The study protocol was approved by the Research Ethics Committee of CEUMA University, São Luís, Maranhão, Brazil (Approval No. 4,945,118), and was prospectively registered in the Brazilian Registry of Clinical Trials (ReBEC; registration No. 11065) on June 30, 2023. All participants provided written informed consent prior to enrollment. Patients were treated at the Dental Clinics of CEUMA University and Florence University. No modifications to the original protocol occurred after the start of patient recruitment and all methods were performed in accordance with the relevant guidelines and regulations.

Eligibility

The inclusion criteria were patients aged 15–55 years with a permanent tooth diagnosed with symptomatic irreversible pulpitis and with sufficient coronal structure to allow rubber dam isolation. All enrolled patients were systemically healthy and classified as American Society of Anesthesiologists physical status I (ASA I). Exclusion criteria were: pregnancy; systemic conditions contraindicating endodontic treatment (e.g., recent myocardial infarction, uncontrolled hypertension, uncontrolled diabetes); intake of any analgesic/anti-inflammatory medication prior to treatment; allergy to anesthetics; internal or external resorption; periodontal disease (probing depth > 3 mm, mobility, endo-periodontal involvement); orthodontic movement of the affected tooth; teeth with anatomical alterations; non-cooperative patients; and intolerance to non-steroidal anti-inflammatory drugs. Diagnosis was established based on the patient's chief complaint, medical and dental history, clinical and radiographic examination, and complementary tests, including cold pulp sensibility test with Endo-Frost spray (Coltene-Whaledent, Langenau, Germany), periodontal probing, apical palpation, and vertical and horizontal percussion.

Sample size

Sample size was calculated for a superiority trial, assuming a 95% confidence level, 80% power, postoperative pain incidence of 34%[18], and an expected success rate of 66% in the experimental treatment. A minimum of 35

patients per group was required. The calculation was performed using Sealed Envelope™ (www.sealedenvelope.com/power).

Randomization and allocation

Patients were randomly assigned to two groups (Laser vs. Control) using block randomization (blocks of 4) with Sealed Envelope™. Allocation concealment was ensured by an independent researcher (C.N.C.), who prepared opaque sealed envelopes containing the treatment assignment. Each envelope was opened immediately before intervention after patient enrollment.

Blinding

Patients were blinded regarding laser use during the emergency endodontic procedure. All patients wore protective infrared safety glasses covered with gauze to prevent awareness of treatment allocation. In the control group, the laser handpiece was positioned identically to the experimental group but remained inactive (sham procedure).

Intervention

All treatments were performed by a single experienced operator. The professional was trained, certified, and calibrated in low-level laser therapy. All laser procedures were standardized and followed the same protocol. Local anesthesia was administered, and the treated tooth was isolated with a rubber dam. Endodontic access was prepared, and complete pulpectomy was performed using Hedström files (Dentsply Maillefer, Ballaigues, Switzerland) under irrigation with 2.5% sodium hypochlorite solution (Fórmula & Ação, São Paulo, Brazil). The apparent tooth length was measured on a periapical radiograph using a plastic ruler. The provisional working length was defined as this length minus 3 mm, to account for potential radiographic distortion. After pulpectomy, the access cavity was temporarily restored, and occlusion was checked. No intracanal medication was applied. In the experimental group, infrared AsGaAl low-level laser therapy (wavelength: 808 nm, power: 100 mW; Photon Lase III, DMC Equipamentos, São Carlos, Brazil) was applied immediately after treatment. The protocol used continuous emission mode, with the fiber tip placed in contact and perpendicular to the dental surface. Up to six irradiation points were applied, depending on tooth type: Molars/premolars: two points at the middle third of the buccal crown; two points at the cemento-enamel junction (buccal aspect); one point at the apex of each buccal root. Anterior teeth: one point at the middle third of the buccal crown; one point at the cemento-enamel junction (buccal aspect); one point at the apex of each buccal root. Each point received 10 s of irradiation (1 J), totaling up to 60 s (60 J). This dosimetry protocol was adopted based on a previous randomized clinical trial with a similar clinical context and outcome assessment[18]and was selected to ensure adequate tissue penetration and energy delivery to inflamed periapical tissues during emergency endodontic procedures. In the control group, the laser handpiece was positioned identically, but the device remained inactive.

Outcome measures

The primary outcome was postoperative pain in permanent teeth diagnosed with symptomatic irreversible pulpitis after emergency pulpectomy. Secondary outcomes included tenderness to biting after treatment. Pain intensity was recorded using two validated scales: the Visual Analogue Scale (VAS, 0–10 cm) and the Numerical Rating Scale (NRS, 0–10). For the VAS, patients marked their perceived pain on a 10-cm horizontal line, categorized as: 0 = no pain; 1–2 = mild pain; 3–4 = moderate pain; 5–7 = considerable pain; 8–10 = severe pain. For the NRS, patients rated pain intensity on a 0–10 line, with the score measured in millimeters from the zero point. Patients recorded spontaneous pain levels at 6, 12-, 24-, 48-, and 72-hours post-treatment. To assess tenderness to biting, patients were instructed to bite a 3 × 2 cm latex device (previously provided by the researchers) at the treated tooth, and pain presence/absence was recorded[30]. No routine medication was prescribed after treatment. Patients were advised to take 600 mg of ibuprofen in cases of moderate pain[31]and to contact the operator if pain persisted.

Data analysis

Pearson's Chi-square test (χ^2) was used to verify whether there were statistically significant differences between the groups regarding the proportions of male and female participants, as well as the need for medication use. Mann-Whitney test was applied to analyze whether there were significant differences between the groups in the mean age, the number of tablets used, and the time until pain cessation. Fisher's Exact test (αF2) was employed to determine whether there were significant differences between the groups in the distribution of tooth type. To evaluate the effect of the type of therapy (Control vs. Laser), the participant's gender, age, and tooth type (Incisor, Canine, Premolar, and Molar) on the degree of pain measured by NRS and VAS at all assessment time points, an ordinal logistic regression using the Generalized Estimating Equations (GEE) method with a robust covariance matrix was applied, while for tenderness to biting, which was assessed as a dichotomous variable (presence or absence of pain), a binary logistic regression also using a GEE model was applied. All analyses were performed using IBM SPSS Statistics for Windows, v.27 (IBM Corp., Armonk, N.Y., USA). In all tests, the significance level adopted was 5%. All randomized participants received the allocated intervention and were included in the primary outcome analysis.

Results

A total of 91 teeth (1 per patient) were initially assessed for eligibility, based on inclusion criteria: patients aged 15–55 years with permanent teeth diagnosed with symptomatic irreversible pulpitis and sufficient coronal structure for rubber dam isolation. Three teeth were excluded at screening due to previous emergency endodontic procedures ($n = 2$) and loss of documentation with clinical results ($n = 1$). Additional exclusions after eligibility

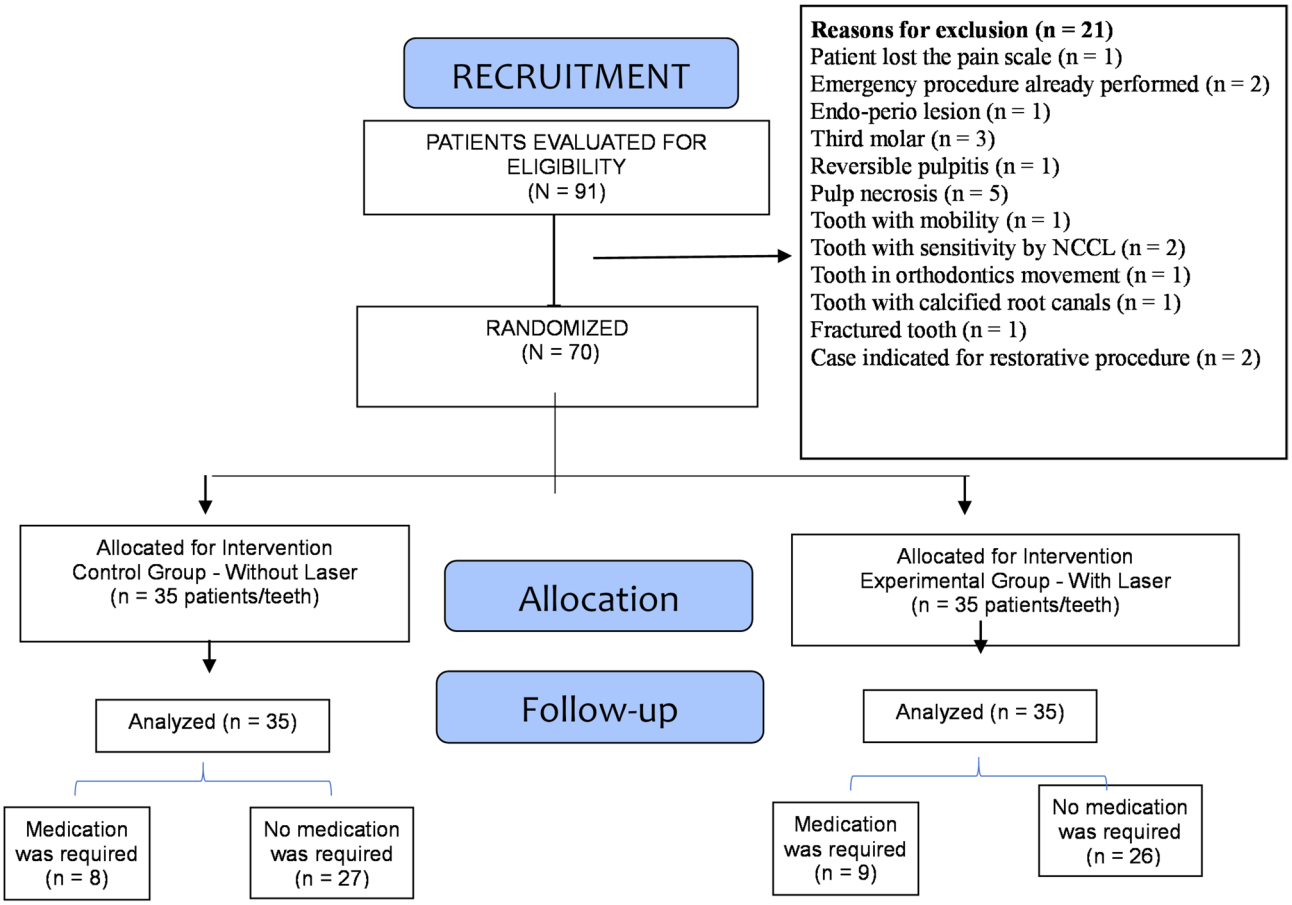


Fig. 1. Flowchart of the clinical trial according to CONSORT.

		Control group	Laser group	*p*-value
Sex	Male	15 (42.9%)	11 (31.4%)	0.322*
	Female	20 (57.1%)	24 (68.6%)	
Age		34.03 (11.80%)	30.91 (11.33%)	0.285**
Tooth Type	Incisive	1 (2.9%)	0 (0.0%)	0.809***
	Canine	1 (2.9%)	1 (2.9%)	
	Molar	18 (51.4%)	21 (60.0%)	
	Premolar	15 (42.9%)	13 (37.1%)	
Medication	No	28 (80.0%)	26 (74.3%)	0.324*
	Yes	7 (20.0%)	9 (25.7%)	
Number of medication tablets		0.80 (1.92)	1.00 (2.18)	0.587**
Pain relief (in days)		0.54 (1.17)	0.50 (1.13)	0.993**

Table 1. Demographic and clinical characteristics, medication use and quantity, and days required for postoperative pain relief ($n = 70$). *Chi-square** Mann-Whitney, *** Fisher`s exact test.

assessment included: endo-perio lesion ($n = 1$), complaint of pain due to partially erupted third molar ($n = 3$), reversible pulpitis ($n = 1$), pulpal necrosis ($n = 5$), tooth with mobility ($n = 1$), sensitivity related to non-carious cervical lesion (NCNL) ($n = 2$), orthodontic movement ($n = 1$), calcified palatal canal ($n = 1$), furcation fracture ($n = 1$), and carious lesions requiring restoration ($n = 2$). A flow diagram of patient recruitment and allocation is shown in Fig. 1. Seventy participants were evaluated, of whom 26 (37.1%) were male and 44 (62.9%) were female. Table 1 presents the frequency distribution according to group and sex of the participants. No significant differences were observed between sex frequencies ($p > 0.05$), indicating that the groups were balanced.

Table 2 shows the mean and standard deviation of postoperative pain intensity measured by the NRS (0–10 cm) and VAS (0–10 cm) scales, as well as the relative frequency (%) of cases of pain (tenderness on

Pain scale	Group	6 h	12 h	24 h	48 h	72 h
NRS	Control	1.51 ± 2.50	1.37 ± 2.32	0.94 ± 1.86	0.66 ± 1.16	0.49 ± 1.01
	Laser	2.03 ± 2.97	2.06 ± 2.86	1.63 ± 2.51	1.43 ± 2.63	1.14 ± 2.03
VAS	Control	1.43 ± 2.21	1.31 ± 2.11	0.83 ± 1.61	0.80 ± 1.41	0.51 ± 0.95
	Laser	2.34 ± 3.04	2.00 ± 2.95	1.89 ± 2.96	1.40 ± 2.83	1.29 ± 0.95
Tenderness on biting	Control	28.6%	31.4%	11.4%	11.4%	11.4%
	Laser	26.5%	35.3%	29.4%	26.5%	20.6%

Table 2. Mean and standard deviation (SD) of postoperative pain intensity, measured by NRS (0–10) and VAS (0–10 cm); and relative frequency (%) of tenderness on biting, according to the assessment times for both study groups.

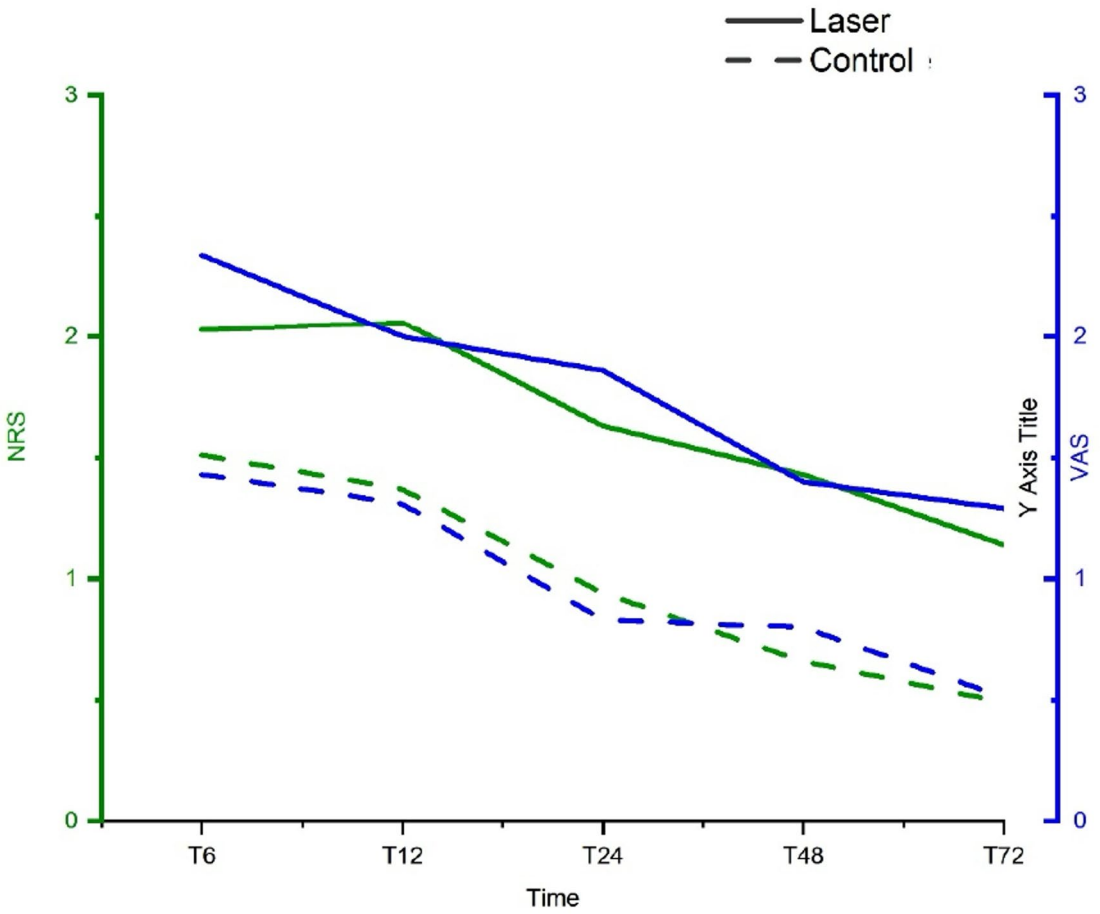


Fig. 2. Pain means values (assessed by NRS and VAS scales) during the assessed times.

biting) assessed by the device for biting (DISP), at the time intervals following the endodontic emergency in both groups. Figure 2 illustrates the trend of mean NRS and VAS pain scores across the evaluation periods. A progressive decrease in scores was observed over time, regardless of the scale used. Although changes in pain scores were observed at specific time points, these differences reflect intra-group variation over time and were not indicative of a significant inter-group treatment effect. The pattern of decrease over time was similar between groups. Statistical analyses demonstrated that there was no significant difference between the Control and Laser groups on any of the pain scales. Only the time factor had a significant effect on pain scores. The results of the longitudinal regression analysis are presented in Table 3.

Discussion

This study evaluated postoperative pain outcomes following emergency endodontic treatment with or without adjunctive photobiomodulation therapy (PBM) in patients diagnosed with symptomatic irreversible pulpitis—the most common chief complaint in this patient group[18]. Low-level laser therapy (LLLT) has been proposed

Pain scale		*p*-value
NRS	Treatment type	0.254
	Sex	0.888
	Tooth	0.285
	Age	0.983
	Time	0.001
VAS	Treatment type	0.524
	Sex	0.586
	Tooth	0.235
	Age	0.552
	Time	0.001
Tenderness on biting	Treatment type	0.273
	Sex	0.983
	Tooth	0.341
	Age	0.275
	Time	0.038

Table 3. Results of the ordinal and binary logistic regression model using generalized estimating equations (GEEs) evaluating the effect of treatment type, sex, tooth type, age, and time factor on pain assessed by different types of pain scales.

as an effective alternative to analgesics and anti-inflammatory drugs because it exerts multiple localized effects, including increased tissue circulation without altering body temperature, removal of pain-inducing substances such as prostaglandins, and inhibition of inflammatory mediator production[18].

The diode laser used in this study has long been applied effectively for orofacial pain control[32]. Its selection was supported by Chow et al.[33], who demonstrated that laser light absorbed by nociceptors induces analgesia by inhibiting A- and C-type nerve fibers, thereby reducing peripheral nerve conduction velocity and ultimately decreasing pain following endodontic procedures.

Although lower fluence values have been reported in other studies, there is currently no consensus regarding an optimal PBM dosimetry protocol for endodontic pain control. Differences in inflammatory burden, target tissue depth, and clinical scenario may influence the required energy dose and partially explain divergent outcomes[34].

In this study, postoperative pain peaked between 12 and 24 h after treatment in both groups, as assessed using the Numerical Rating Scale (NRS) for spontaneous pain. A systematic review and meta-analysis reported that approximately 40% of patients experience moderate pain 24 h after endodontic treatment[35]. However, despite the reported benefits of LLLT in previous research, its superiority for postoperative pain control was not demonstrated in the present study.

Some patients in both the control and experimental groups required analgesic medication. Ibuprofen 600 mg was selected as the rescue drug because, among nonsteroidal anti-inflammatory drugs (NSAIDs), it provides rapid analgesia with minimal adverse effects. Nonetheless, previous studies have described a "ceiling effect," indicating that ibuprofen alone may not achieve optimal pain control and may require combination therapy for enhanced efficacy[18].

Limitations of this clinical study include variability in preoperative conditions of the treated teeth, such as anatomical differences, and the inherent subjectivity of pain perception. To minimize bias, two pain assessment tools were used: the Visual Analogue Scale (VAS) and the Numerical Rating Scale (NRS)[9,36]. According to Williamson and Hoggart[37], the numerical scale is easier to interpret and offers greater statistical sensitivity. However, due to the high heterogeneity of subjective pain experiences, the combined use of both scales provides more reliable data. The age range of participants in the present study was relatively broad (15–55 years), and teeth from both maxillary and mandibular arches were included. Although age and tooth location may influence postoperative pain perception, randomization ensured a balanced distribution of these variables between the Control and LLLT groups, minimizing potential confounding effects. In addition, age was not significantly associated with postoperative pain outcomes in the adjusted analysis.

The inclusion of different age groups and tooth locations reflects the clinical heterogeneity commonly encountered in emergency endodontic care. Future studies with larger samples may benefit from stratified designs to further explore potential subgroup-specific effects. The absence of standardized preoperative assessment of key clinical variables—including preoperative pain intensity, intrapulpal bleeding time, and extent of carious tissue—limits baseline control and may influence postoperative pain outcomes. In the present study, these parameters were not quantitatively standardized or stratified, reflecting the emergency clinical context in which rapid pain relief and patient comfort were prioritized. Although many cases presented with limited carious exposure—a common finding in teeth diagnosed with symptomatic irreversible pulpitis, in which pain severity is not necessarily proportional to lesion depth—caries extent was not used as a selection or stratification criterion. This lack of preoperative standardization reduces internal validity and restricts the external validity of the findings. Accordingly, the present study should be interpreted as an exploratory, pragmatic investigation conducted under emergency endodontic care conditions, rather than as a definitive efficacy trial.

Although no statistically significant differences were found between groups, a higher proportion of emergency treatments was performed in women than in men. Previous studies have reported that women tend to show greater pain sensitivity to pressure, thermal, and mechanical stimuli[38]. Hashmi[39]also observed that while women generally experience higher pain intensity, they tend to adapt more effectively to painful stimuli.

Nunes et al.[18]compared the effects of LLLT and ibuprofen (600 mg) in 70 patients after endodontic treatment and found that LLLT significantly reduced pain during the first 24 h compared with drug therapy. Recent systematic reviews have also confirmed the potential of PBM therapy in reducing postoperative pain after endodontic procedures. However, variations in application protocols across studies emphasize the need to standardize irradiation parameters, as exposure time appears to play a critical role in treatment success[18,30].

In the present study, occlusal trauma was carefully controlled by using carbon paper to adjust provisional restorations, ensuring that premature contacts did not influence postoperative pain outcomes. Based on the obtained results, the alternative hypothesis—that PBM therapy is effective in reducing or eliminating postoperative pain—was rejected.

In this study, no patients were involved in defining the research questions or outcome measures, nor were they involved in the development of recruitment procedures, study design, or implementation. Likewise, no patients were consulted for the interpretation or reporting of results. This was due to the methodological characteristics of the project, in which all procedures, outcomes, and operational steps were predetermined based on established clinical and laboratory protocols, leaving no scope for patient or public involvement during the design phase.

Conclusion

Adjunctive low-level laser therapy was not associated with additional postoperative analgesic benefit following emergency pulpectomy in permanent teeth with symptomatic irreversible pulpitis.

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author, Dr. Ceci Nunes Carvalho, upon reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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